

Organ-Aware Routing for Robust Medical Image Segmentation: A Failure-Detectable Alternative to Mixed Models

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Abstract—Heterogeneous medical image segmentation pipelines that use a single shared model across anatomically distinct organs sacrifice organ-specific feature learning and degrade silently under distribution shift, offering no signal that predictions have become unreliable. We propose an organ-aware segmentation framework where an upstream classifier routes each input to a dedicated specialist network. Through controlled experiments on chest X-ray pneumothorax and brain MRI tumor datasets, we show that organ-aware routing yields substantially higher segmentation accuracy (+8–14 Dice points) under both clean and noisy conditions compared to a mixed-organ baseline. Furthermore, when routing fails, the organ-aware system exhibits sharp performance collapse accompanied by elevated uncertainty and anomalous plausibility signatures, enabling post-hoc detection via Monte Carlo dropout and morphological checks. The mixed model, by contrast, degrades more gradually but without detectable warning signals. Our results demonstrate that organ-aware pipelines offer both superior accuracy and the ability to *fail loudly*, enabling safer deployment through uncertainty-driven abstention.

Index Terms—medical image segmentation, organ-aware routing, failure detection, uncertainty estimation, robustness

I. INTRODUCTION

Medical image segmentation increasingly relies on deep learning, yet deploying a single model across anatomically heterogeneous inputs (e.g., chest X-rays and brain MRI) introduces a fundamental tension. A mixed model S^H trained jointly on multiple organs avoids explicit routing but cannot specialize its learned representations for each anatomy. Under distribution shift (noise, domain gap), such models degrade *silently*: Dice scores drop without any accompanying signal that predictions are unreliable [1], [2].

Organ-specific specialization offers an alternative. An organ-aware system S^O employs an upstream classifier to route each input to a dedicated specialist network, enabling anatomy-specific feature learning. This improves baseline performance, yet introduces a new failure mode: if the router misclassifies the input organ, the specialist receives out-of-distribution data, causing severe segmentation failure [3].

This raises a critical question: *is a sharp but detectable failure preferable to a gradual but silent one?* Prior work on medical segmentation robustness focuses primarily on accu-

racy metrics under corruption [4], [5], with limited analysis of (i) routing-induced catastrophic failure, (ii) whether such failures are detectable at inference time, and (iii) practical safety mechanisms for deployment.

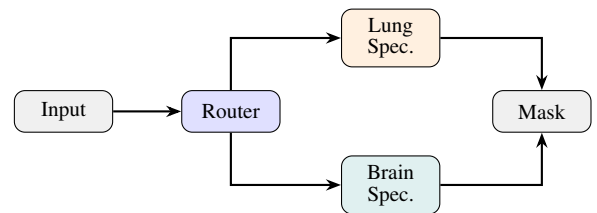
We make three contributions:

- 1) An organ-aware segmentation pipeline with controlled routing analysis across clean, noisy, and misrouted conditions.
- 2) A robustness evaluation showing that specialization improves both accuracy and noise resilience under correct routing while producing sharp, *detectable* failure under misrouting, in contrast to mixed models that degrade silently.
- 3) A post-hoc failure-aware inference mechanism combining MC Dropout uncertainty estimation with morphological plausibility checks, enabling deployment-time abstention.

II. METHOD OVERVIEW

A. Systems Compared

We compare two segmentation architectures. The **mixed model** S^H is a single UNet++ trained jointly on both lung (pneumothorax) and brain (LGG tumor) data with no routing. The **organ-aware system** S^O consists of an organ classifier followed by dedicated specialist UNet++ networks, one per anatomy. Both systems share the same backbone family and loss function to isolate the effect of routing.



III. EXPERIMENTAL SETUP

Datasets. Lung segmentation uses the SIIM-ACR Pneumothorax chest X-ray dataset [10] (12,047 images with binary

TABLE I
EVALUATION SCENARIOS PER ORGAN DOMAIN.

Case	System	Routing	Input
1	S^O (specialist)	Correct	Clean
2	S^H (mixed)	—	Clean
3	S^O (specialist)	Correct	Noisy
4	S^H (mixed)	—	Noisy
5	S^O (wrong spec.)	Forced wrong	Noisy
6	S^O (wrong spec.)	Forced wrong	Clean

masks). Brain segmentation uses the LGG FLAIR MRI segmentation dataset [11] (3,929 slices across 110 patients). Both are split into 70/15/15% train/val/test partitions with stratified sampling by positive-mask ratio.

Architecture and training. All models use UNet++ with EfficientNet encoders and BCE + Dice loss. The specialists employ SCSE attention and 0.1 decoder dropout; the mixed model uses no attention. Training uses AdamW with cosine-annealing learning rate scheduling and mixed-precision (FP16). Conservative augmentations (flips, mild rotations, brightness/contrast) are applied uniformly. Full architectural and optimization details are in the supplementary material (Appendices A–B).

Noise injection. Gaussian noise ($\sigma \in [0.05, 0.06]$) is added in normalized pixel space at test time to simulate mild acquisition perturbation. This is applied identically across all models.

IV. ORGAN-AWARE ROBUSTNESS ANALYSIS

A. Evaluation Scenarios

We evaluate six scenarios per organ domain, as summarized in Table I. Cases 1–4 represent operational conditions; Cases 5–6 are ablations simulating misrouting under noisy and clean inputs respectively. Although our organ classifier achieves near-perfect accuracy in the binary lung-vs-brain setting, we include clean misrouting (Case 6) to study failure detectability under conditions where the routing error is not compounded by input noise. This is directly relevant to more challenging deployment scenarios, such as routing among disease-specific specialists within the same organ (e.g., pneumothorax vs. pleural effusion in chest X-rays), where classifier confusion is far more likely even on clean inputs.

B. Lung (Pneumothorax) Results

Table II reports Dice and IoU on the lung test set ($n=1000$, stratified). The pneumothorax specialist (S^O) achieves 0.837 clean Dice versus 0.694 for the mixed model, a +14.3 point advantage. Under noise, S^O drops to 0.711 (−12.6 pts) while S^H degrades to 0.657 (−3.7 pts from its lower baseline). Under forced misrouting (brain specialist on lung data), Dice collapses to 0.046 (noisy) and 0.155 (clean), confirming catastrophic failure regardless of input quality.

C. Brain (LGG Tumor) Results

Table III reports brain MRI results ($n=590$). The specialist achieves 0.930 clean Dice versus 0.850 for S^H (+8.0 pts). Noise reduces S^O only modestly to 0.905 (−2.5 pts), whereas

TABLE II
LUNG SEGMENTATION RESULTS. FLAG = FRACTION FLAGGED UNRELIABLE.

Scenario	Dice	IoU	Flag
S^O clean (Case 1)	0.837	0.794	0.191
S^H clean (Case 2)	0.694	0.663	0.179
S^O noisy (Case 3)	0.711	0.683	0.226
S^H noisy (Case 4)	0.657	0.635	0.157
S^O misrouted noisy (Case 5)	0.046	0.046	0.971
S^O misrouted clean (Case 6)	0.155	0.155	0.929

TABLE III
BRAIN SEGMENTATION RESULTS. FLAG = FRACTION FLAGGED UNRELIABLE.

Scenario	Dice	IoU	Flag
S^O clean (Case 1)	0.930	0.904	0.219
S^H clean (Case 2)	0.850	0.815	0.225
S^O noisy (Case 3)	0.905	0.878	0.208
S^H noisy (Case 4)	0.837	0.803	0.234
S^O misrouted noisy (Case 5)	0.534	0.534	0.620
S^O misrouted clean (Case 6)	0.529	0.529	0.742

S^H drops to 0.837 (−1.3 pts). Misrouting (pneumothorax specialist on brain data) causes Dice to fall to 0.534 (noisy) and 0.529 (clean).

D. Interpretation

Three findings emerge. First, **specialization improves both accuracy and robustness under correct routing**: the organ-aware system consistently outperforms the mixed model on both domains under both clean and noisy conditions, with Dice gains of +8.0 (brain) and +14.3 (lung) on clean data. Second, **misrouting causes sharp performance collapse**: Dice drops by 40–79 points when the wrong specialist is applied, regardless of whether the input is clean or noisy, compared to 4–13 points of noise-induced degradation in the mixed model. Third, **the mixed model degrades more gradually but without any alarm**; its moderate decline produces no signal distinguishable from normal operation. This asymmetry between sharp-detectable and gradual-silent failure motivates the failure-aware inference framework described next.

V. FAILURE-AWARE INFERENCE

Dice collapse is not observable at deployment time since ground truth masks are unavailable. We therefore design post-hoc detection mechanisms that flag unreliable predictions using only model outputs.

A. Uncertainty Estimation

We employ Monte Carlo (MC) Dropout [6] for models containing dropout layers. At inference, dropout is kept active for $N=10$ stochastic forward passes per image. For each pixel, we compute the variance of the predicted probability across passes. The per-image uncertainty score is the mean variance over foreground pixels (those with mean predicted probability > 0.05). This foreground-weighted formulation prevents large empty backgrounds from diluting the uncertainty signal.

B. Plausibility Checks

We apply simple morphological heuristics to flag anatomically implausible masks: (i) *area ratio*: the predicted mask area as a fraction of total image area is checked against organ-specific minimum thresholds; (ii) *connected components*: the mask fragment count is compared to organ-specific maximums (e.g., ≤ 5 for brain, ≤ 30 for lung). An image is flagged if uncertainty exceeds a calibrated threshold *or* any plausibility check fails.

C. Detectability Analysis

The flag rates reported in Tables II and III combine uncertainty estimation and domain shift detection into a single reliability indicator. Uncertainty thresholds are calibrated at the 90th percentile of clean-correct operating variance (frozen before evaluation). An image is flagged if *either* signal exceeds its threshold.

The key result is the **asymmetry in detectability**. Misrouted cases produce near-total flagging (92.9–97.1% for lung, 62.0–74.2% for brain), reflecting the model’s recognition that it is processing out-of-distribution anatomy. Notably, clean misrouting (Case 6) is also reliably detected because the domain shift signal alone is sufficient even without noise amplification. By contrast, the mixed model under noise (Case 4) shows *lower* flag rates (15.7% lung, 23.4% brain) than even clean-correct operation, meaning its degradation is entirely invisible to the monitoring system.

Two complementary signals drive this detection. *Uncertainty flagging* captures epistemic confusion via MC Dropout variance: misrouted lung cases trigger 48.2% uncertainty flags versus 9.0% under correct clean routing. *Domain shift flagging* via encoder feature cosine distance detects anatomical mismatch: misrouted lung images produce a mean shift score of 0.953 against a nominal centroid of 0.539. Together, these mechanisms provide redundant coverage; even when one signal is ambiguous, the other typically fires. This validates our central claim: misrouting produces abnormal signatures that enable failure detection fundamentally absent in mixed models. The organ-aware system fails loudly; the mixed model fails silently.

VI. DISCUSSION

A. Accuracy vs. Detectability Trade-off

The results reveal a fundamental design trade-off between *graceful degradation* and *detectable failure*. The mixed model never collapses catastrophically (its worst-case Dice of 0.657 on noisy lung is still within operational range), but its errors are indistinguishable from normal variance. The organ-aware system achieves higher peak accuracy (+8–14 Dice points) and maintains robustness under noise, but misrouting produces sharp collapse. Critically, this collapse is accompanied by anomalous uncertainty signatures that enable automatic detection.

This trade-off has a clear resolution in safety-critical settings: a **detectable failure that can be intercepted** is strictly preferable to an **undetectable one that propagates silently**

into clinical decisions. The organ-aware system provides this detectability by design.

B. Deployment Implications

In clinical deployment, the organ-aware system benefits from a layered safety strategy. First, high-confidence routing ($>95\%$ classifier accuracy) ensures misrouting is rare. Second, when misrouting does occur, the uncertainty and domain shift signals enable automatic flagging. Third, flagged predictions can trigger one of several fallback actions: (i) defer to the mixed model as a safe but lower-accuracy alternative, (ii) route to human review, or (iii) abstain entirely with an explicit “unreliable” warning.

The system-level Dice after abstention (replacing flagged predictions with zero) demonstrates this value: under misrouting, the system Dice drops to near zero (0.000–0.019 lung, 0.183–0.270 brain), but 93–97% (lung) and 62–74% (brain) of these catastrophic predictions are *correctly identified* for rejection, substantially limiting downstream harm. In the mixed model, no analogous rejection mechanism exists because the uncertainty profile is indistinguishable from normal operation.

C. Limitations

This study has several limitations. First, we evaluate a binary organ setting (lung vs. brain); scaling to multi-organ routing with $k > 2$ specialists remains untested and may introduce more complex failure modes. In particular, routing among disease-specific specialists within the same organ (e.g., different lung pathologies) would face higher classifier confusion rates, making the clean-misroute analysis (Case 6) especially relevant as a proxy for such settings. Second, our noise model uses synthetic Gaussian perturbation ($\sigma \in [0.05, 0.06]$) in normalized space; realistic acquisition artifacts (motion blur, coil inhomogeneity, metal artifacts) may produce different uncertainty signatures. Third, the uncertainty thresholds are calibrated at the 90th percentile of clean-correct variance and may require site-specific recalibration for new scanners or protocols. Finally, the morphological plausibility checks (area ratio, connected components) are organ-specific heuristics that require manual tuning per anatomy.

VII. CONCLUSION

We demonstrated that organ-aware routing improves segmentation accuracy over mixed models under both clean and noisy conditions, with gains of +8.0 Dice points (brain) and +14.3 points (lung) on clean data. When routing errors occur, the resulting performance collapse is sharp but accompanied by elevated uncertainty and anomalous morphological signatures, enabling reliable post-hoc failure detection with flag rates of 62–97%. Mixed models, by contrast, degrade silently with flag rates indistinguishable from normal operation ($\leq 23.4\%$). These results establish that organ-aware systems not only improve accuracy but also *fail loudly*, enabling uncertainty-driven abstention and safer deployment of heterogeneous medical image segmentation pipelines.

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APPENDIX

A. Architecture Details

All segmentation models use the UNet++ decoder [2] with pre-trained EfficientNet encoders [7]. Table IV summarizes the architectural differences.

TABLE IV
MODEL ARCHITECTURE SUMMARY.

Model	Encoder	In Ch	Attention	Dropout
Brain spec.	EfficientNet-B3	1	SCSE	0.1
Lung spec.	EfficientNet-B1	3	SCSE	0.1
Mixed S^H	EfficientNet-B1	3	None	0.1

The brain specialist uses a single-channel input (grayscale FLAIR) with EfficientNet-B3 (12M params). The lung specialist and mixed model both use 3-channel input (grayscale replicated) with EfficientNet-B1 (7.8M params). SCSE attention [8] is used in both specialists for spatial-channel feature recalibration. The mixed model omits attention to represent a simpler single-model baseline. All models output single-channel logit maps thresholded at 0.5.

B. Training Hyperparameters

Table V lists all training hyperparameters.

TABLE V
TRAINING HYPERPARAMETERS.

Parameter	Brain	Lung	Mixed
Optimizer	AdamW	AdamW	AdamW
Learning rate	1×10^{-4}	3×10^{-4}	1×10^{-4}
Weight decay	1×10^{-5}	1×10^{-5}	1×10^{-5}
Batch size	24	8 (eff. 16)	46
Max epochs	45	45	30
Early stopping	10 ep.	15 ep.	15 ep.
Loss	BCE + $1.0 \times \text{Dice}$	BCE + $1.5 \times \text{Dice}$	BCE + $1.0 \times \text{Dice}$
Image size	256	512	256
Precision	FP16	FP16	FP16
Seed	42	42	42

Augmentations (all models): horizontal flip ($p=0.5$), rotation ($\pm 10^\circ$, $p=0.4$), shift-scale-rotate (shift 5%, scale 5%, $p=0.4$), brightness/contrast ($\pm 15\%$, $p=0.3$), Gaussian noise ($\sigma \in [10, 30]$, $p=0.2$). No elastic deformations. Validation uses no augmentation.

Normalization: Specialists use fixed normalization ($\mu=0.5, \sigma=0.5$) mapping to $[-1, 1]$. The mixed model uses per-image z-score normalization.

C. Extended Metrics

Table VI reports additional metrics for each scenario.

D. Threshold Sensitivity

Uncertainty thresholds are calibrated at the p -90 of clean-correct operating variance. Varying this percentile trades off false flag rate against misrouting detection sensitivity. At p -80, flag rates increase $\sim 5\%$ across all cases, improving misrouting recall at the cost of more abstentions under correct operation.

TABLE VI

EXTENDED EVALUATION METRICS. DS = DOMAIN SHIFT SCORE (COSINE DISTANCE FROM CALIBRATED CENTROID). UNC FLAG = FRACTION FLAGGED BY UNCERTAINTY ALONE. SHIFT FLAG = FRACTION FLAGGED BY DOMAIN SHIFT ALONE.

	Scenario	Dice	IoU	DS	Unc	Unc%	Shift%
Lung	S^O clean	.837	.794	.539	.022	9.0	13.5
	S^H clean	.694	.663	.286	.025	8.7	10.9
	S^O noisy	.711	.683	.446	.031	19.3	5.1
	S^H noisy	.657	.635	.275	.024	9.6	7.2
	S^O misrt. noisy	.046	.046	.953	.047	48.2	91.9
	S^O misrt. clean	.155	.155	.931	.034	43.8	88.3
Brain	S^O clean	.930	.904	.741	.006	9.8	12.7
	S^H clean	.850	.815	.474	.017	9.0	13.1
	S^O noisy	.905	.878	.632	.009	12.7	8.3
	S^H noisy	.837	.803	.467	.019	11.2	11.5
	S^O misrt. noisy	.534	.534	.728	.059	39.7	27.6
	S^O misrt. clean	.529	.529	.806	.057	38.6	49.0

At p -95, misrouting detection drops below 90% for lung but remains above 95% due to the extreme domain shift signal. The domain shift threshold (p -90 of pooled clean+noisy cosine distance) is robust to ± 5 percentile variation.

E. Additional Qualitative Observations

Under correct routing, specialist predictions closely match ground truth boundaries with smooth, anatomically plausible contours. Misrouted predictions exhibit hallucinated structures: the brain specialist applied to lung images produces small scattered fragments unrelated to pneumothorax anatomy, while the lung specialist applied to brain MRI generates diffuse over-segmentation covering non-tumor regions. These morphological anomalies are captured by the plausibility checks (excess connected components, implausible area ratios) and complement the uncertainty signal for robust failure detection.